

Expression: counts, normalisation, and differential expression

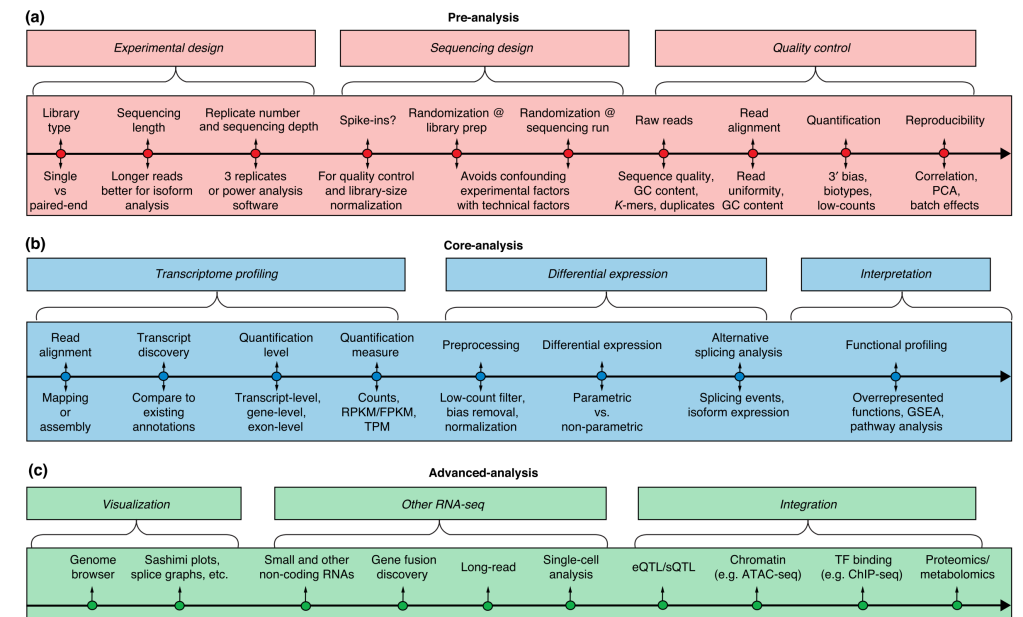
- İfade: sayımlar, normalizasyon ve farklı ifade analizi
- BIB Week 7

Ad of the Week #7

- Skills Ledger (Beceri Defteri): tick quantification and differential expression

One read, from the flow cell to a cell in a matrix

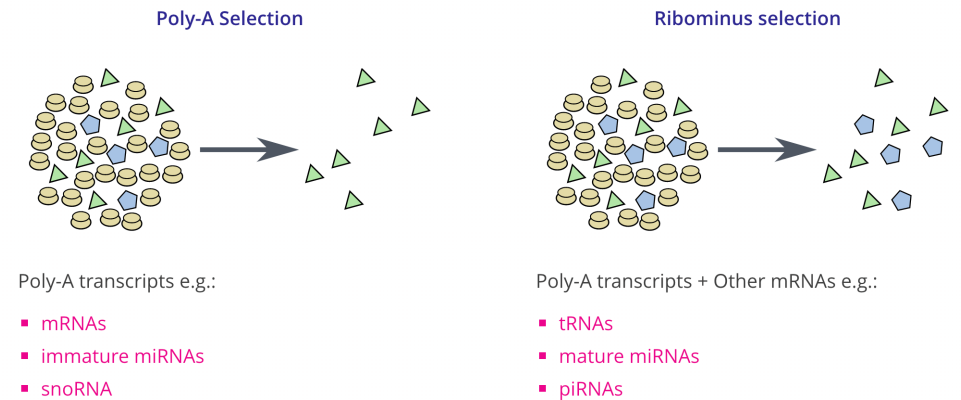
- The read exists. Everything after it is a decision somebody made.
- Decision 1: which molecules got into the library (kütüphane) at all
- Decision 2: which reference and which annotation release (anotasyon sürümü)
- Decision 3: which tool assigned the read to a gene
- Decision 4: what happened to it when it matched two genes
- The number in your matrix is the residue of all four



RNA-seq analysis overview — Conesa et al., Genome Biol 2016;17:13, Fig. 1 (open access)

Decision 1: what got into the library

- rRNA is ~80–90% of total cellular RNA; mRNA is only ~1–5%
- polyA selection (oligo-dT pulldown): cheap, clean, standard for mRNA
 - misses histone mRNAs, many lncRNAs, circRNAs, enhancer RNAs, most bacterial RNA
- rRNA depletion (Ribo-Zero / RiboCop): keeps non-polyadenylated RNA and degraded/FFPE input
 - costs more depth; carries intronic and pre-mRNA signal
- Stranded (yöne özgü) vs unstranded: stranded resolves antisense and overlapping gene pairs
- If nobody can tell you which of these a count matrix is, you cannot interpret it



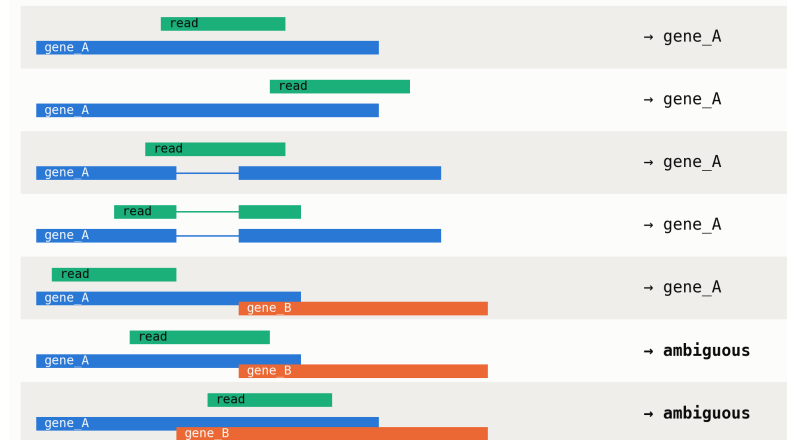
Poly-A selection vs ribo-depletion — what each keeps (CRUK Cambridge Institute)

Decision 2 and 3: reference, annotation, and who counts

- featureCounts / HTSeq + a GTF: align to the genome, then count reads over annotated exons
 - Liao, Smyth & Shi, Bioinformatics 2014;30:923
- salmon / kallisto + a transcriptome FASTA: no genome alignment, probabilistic assignment
 - Patro et al., Nat Methods 2017;14:417 · Bray et al., Nat Biotechnol 2016;34:525
- Both are defensible. They do not give the same numbers.
- The GTF and the FASTA must be the same annotation release — and you must name it

Which gene does the read count for? (featureCounts, default settings)

intron = thin line · spliced read = two pieces · the read must overlap exactly one gene model to be counted



Redrawn in the course style after the hbctraining DGE workshop counting cartoon (https://hbctraining.github.io/DGE_workshop/). Ambiguous reads are discarded unless -O / -M is set.

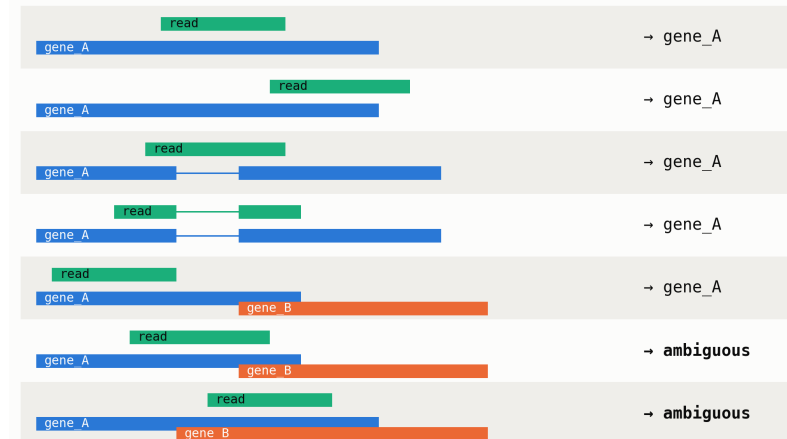
How featureCounts assigns a read to a gene — seven cases, two of them ambiguous (redrawn after hbctraining)

Decision 4: the read that matched two genes

- ~10–20% of RNA-seq reads are multi-mapping (çoklu eşleşen okuma) in human
- featureCounts discards them by default (-M to keep)
- salmon/kallisto distribute them probabilistically across transcripts
- Parologue families, pseudogenes and rRNA loci are where this bites
- Same reads, same genome, two defensible answers, different matrices

Which gene does the read count for? (featureCounts, default settings)

intron = thin line · spliced read = two pieces · the read must overlap exactly one gene model to be counted



Redrawn in the course style after the hbc training DGE workshop counting cartoon (https://hbctraining.github.io/DGE_workshop/). Ambiguous reads are discarded unless -O / -M is set.

The two 'ambiguous' rows: discard (featureCounts default) or distribute (salmon/kallisto)

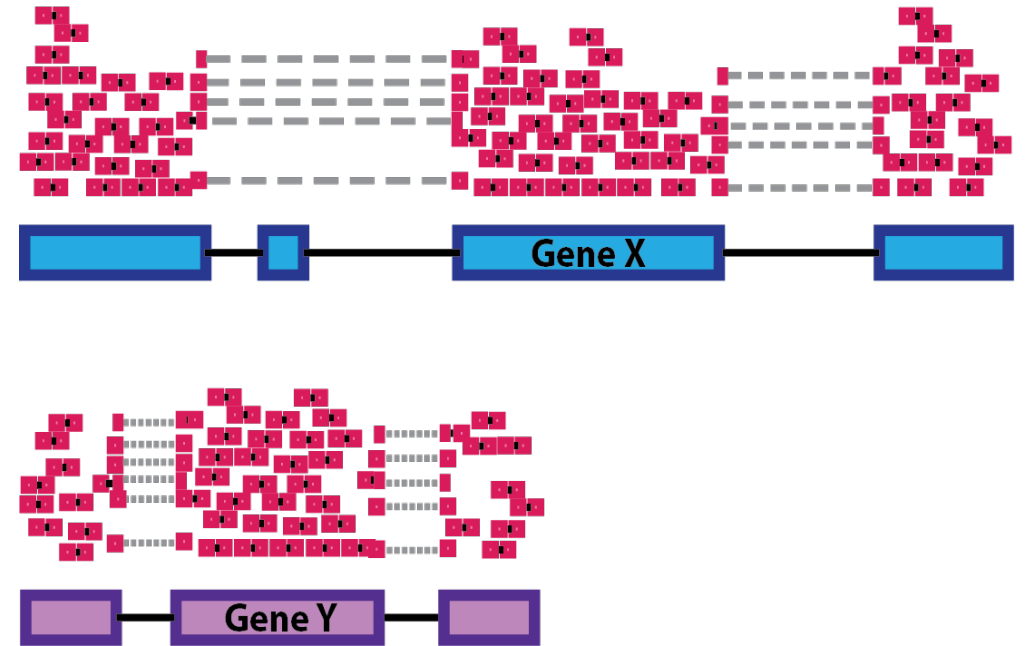
The annotation release is a variable, not a constant — GENOMES G

- The same gene changes count between GENCODE releases: exons are added, models are merged, genes are retired
- ~19,900 protein-coding genes, ~62,000 total gene loci — the exact figure changes every release
- The old deck's "25,000–30,000 genes" is retired from this course as of Week 2
- Zhao & Zhang, BMC Genomics 2015;16:97 — annotation choice measurably changes quantification
- Your GENOMES G row today names the release used for gene lengths

Three unwanted variables stand between counts and comparison

- Sequencing depth (dizileme derinliği) — a sample with twice the reads has twice the counts, everywhere
- Gene length (gen uzunluğu) — a 10 kb transcript catches more fragments than a 1 kb one at equal molarity
- Library composition (kütüphane bileşimi) — a few very highly expressed genes eat the measurement budget
- Every normalisation method you will ever meet corrects a subset of these three
- Which subset it corrects tells you what comparison it licenses

Sample A Reads



Length bias: a longer gene catches more fragments at equal molarity (hbctraining)

The table you keep for the rest of your career

- CPM — counts scaled by total reads → depth only
- TPM — per-kb then rescaled to a million → depth + length
- RPKM / FPKM — per million then per kb → depth + length
- DESeq2 median-of-ratios (ölçek katsayısı) → depth + composition
- edgeR TMM → depth + composition (+ length in the RPKM variant)
- Between-sample comparison needs the composition column. TPM and FPKM do not have it.

What each normalisation actually corrects

What each normalisation corrects — and therefore what it licenses

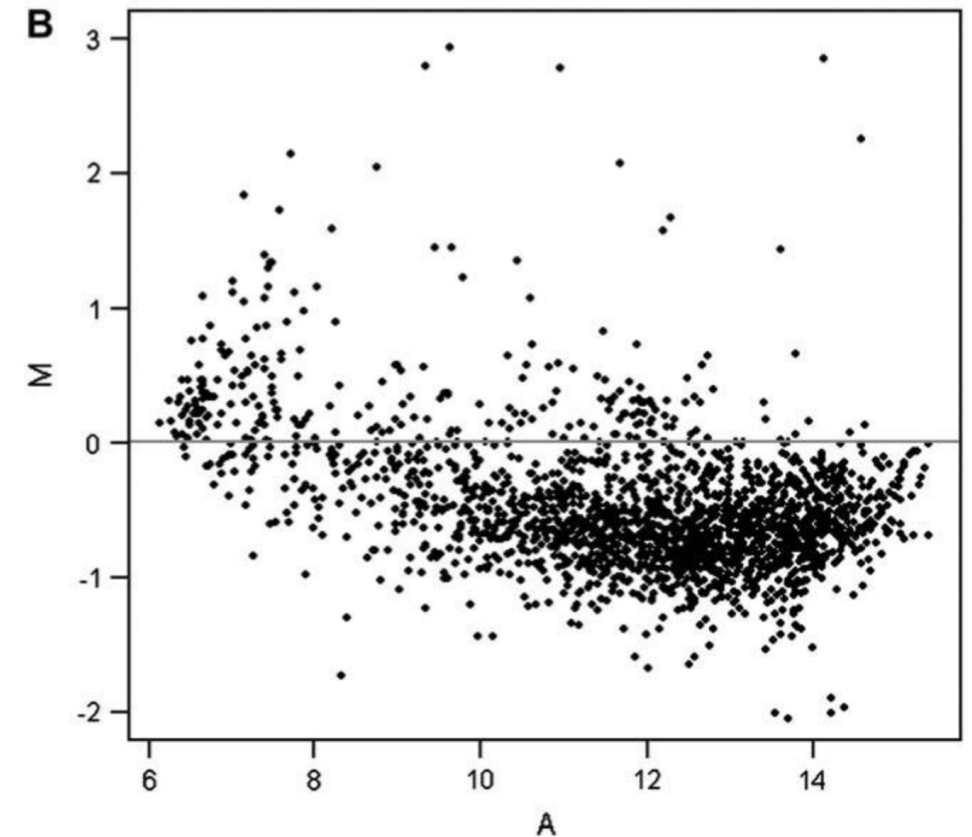
method	depth	length	composition	comparison it licenses
CPM	✓	–	–	replicates of one group; 3'-tag data
TPM	✓	✓	–	within a sample, or replicates of one group — NOT for differential expression
RPKM / FPKM	✓	✓	–	within a sample (legacy) — NOT for differential expression
DESeq2 median-of-ratios	✓	–	✓	between samples: differential expression
edgeR TMM	✓	–	✓	between samples: differential expression

Only the composition column licenses a between-sample comparison. TPM and FPKM do not have it.
TMM also has a length-aware RPKM variant. After Dillies et al. 2013 and Evans et al. 2018 (reviews); redrawn from the 2025 L10 table.

What each normalisation corrects — depth, length, composition — and the comparison it licenses

Five minutes of history, because the idea outlived the platform

- Microarrays needed normalisation before sequencing existed — the problem is not new
- RMA and LOESS — named once here, and dropped
 - Irizarry et al., Biostatistics 2003;4:249 · Yang et al., NAR 2002;30:e15
- What survives: composition bias is real, and it predates sequencing
- What also survives: the MA plot, which you will meet again in twenty minutes
- The marray Bioconductor PDFs are off this course's reading list



Two-colour microarray MA plot: $M = \log(R/G)$ against $A = \text{mean log intensity}$ — the plot outlived the platform

The board: three genes, two samples, real numbers

- Lengths: A = 1 kb, B = 1 kb, C = 10 kb
- Sample 1 counts: A = 100, B = 100, C = 800 $\rightarrow N_1 = 1,000$
- Sample 2 counts: A = 100, B = 100, C = 1,800 $\rightarrow N_2 = 2,000$
- Nothing changed for A or B. Only C went up, and the library got bigger.
- Two recipes, same data. Predict before we compute.

gene	length	sample 1	sample 2
A	1 kb	100	100
B	1 kb	100	100
C	10 kb	800	1,800
N (total reads)		1,000	2,000

Nothing changed for A or B. Only C went up, and the library got bigger.

Predict before we compute: did gene A go up, go down, or stay the same?

The board: three genes, two samples, lengths and raw counts ($N_1 = 1,000$, $N_2 = 2,000$)

FPKM: divide by library size, then by length

- $\text{FPKM} = \text{count} \div (\text{length in kb}) \div (N \div 10^6)$
- Sample 1 ($N_1/10^6 = 0.001$): $A = 100,000 \cdot B = 100,000 \cdot C = 80,000 \rightarrow$ column sum 280,000
- Sample 2 ($N_2/10^6 = 0.002$): $A = 50,000 \cdot B = 50,000 \cdot C = 90,000 \rightarrow$ column sum 190,000
- Gene A: $100,000 \rightarrow 50,000$. FPKM says A halved.
- The two column sums are different. Remember that.

$$\text{FPKM} = \text{count} \div \text{length (kb)} \div (N \div 10^6)$$

gene	length	count s1	count s2	FPKM s1	FPKM s2
A	1 kb	100	100	100,000	50,000
B	1 kb	100	100	100,000	50,000
C	10 kb	800	1,800	80,000	90,000
$N \div 10^6$		0.001	0.002		
column sum				280,000	190,000

Gene A: $100,000 \rightarrow 50,000$. FPKM says A halved — its count never moved.

The two column sums are different. Nothing forces them to be equal. Remember that.

FPKM on the board: $A = 100,000 \rightarrow 50,000$; column sums 280,000 and 190,000

TPM: divide by length FIRST, then rescale to a million

- $\text{rate} = \text{count} \div \text{length(kb)}$. Sample 1 rates: 100, 100, 80 $\rightarrow \Sigma = 280$
- $\text{TPM}_1 = \text{rate} \div 280 \times 10^6 \rightarrow A = 357,143 \cdot B = 357,143 \cdot C = 285,714 \rightarrow \Sigma = 1,000,000$
- Sample 2 rates: 100, 100, 180 $\rightarrow \Sigma = 380$
- $\text{TPM}_2 \rightarrow A = 263,158 \cdot B = 263,158 \cdot C = 473,684 \rightarrow \Sigma = 1,000,000$
- TPM says A fell by ~26%. FPKM said 50%. Same counts.
- Every TPM column sums to exactly 10^6 . That is not a coincidence, it is the definition.

$\text{rate} = \text{count} \div \text{length (kb)} \rightarrow \text{TPM} = \text{rate} \div \Sigma \text{ rates} \times 10^6$ (length FIRST, then rescale)

gene	length	count s1	count s2	rate s1	rate s2	TPM s1	TPM s2
A	1 kb	100	100	100	100	357,143	263,158
B	1 kb	100	100	100	100	357,143	263,158
C	10 kb	800	1,800	80	180	285,714	473,684
column sum		1,000	2,000	280	380	1,000,000	1,000,000

the invariant

TPM says A fell by ~26 % (357,143 $\rightarrow$ 263,158). FPKM said 50 %. Same counts.

Every TPM column sums to exactly 10^6 . Not a coincidence — the definition.

TPM on the board: length first, then rescale — both columns sum to exactly 1,000,000

The swap: same data, two recipes, opposite conclusions

- Compare gene A in sample 1 against gene C in sample 2
- By FPKM: 100,000 vs 90,000 → A is higher
- By TPM: 357,143 vs 473,684 → C is higher
- The ranking reversed, and no data changed
- Within one sample TPM is just FPKM rescaled — ranks are safe. Across samples they are not.
- Wagner, Kin & Lynch, Theory Biosci 2012;131:281 — "the RPKM measure is inconsistent among samples"

gene	length	FPKM s1	FPKM s2	TPM s1	TPM s2
A	1 kb	100,000	50,000	357,143	263,158
B	1 kb	100,000	50,000	357,143	263,158
C	10 kb	80,000	90,000	285,714	473,684
column sum		280,000	190,000	1,000,000	1,000,000

Compare gene A in sample 1 with gene C in sample 2 — same counts, two recipes:

FPKM: 100,000 vs 90,000 → A is higher

TPM: 357,143 vs 473,684 → C is higher

The ranking reversed and no data changed. Within one column ranks are safe; across samples they are not.

Wagner, Kin & Lynch, Theory Biosci 2012;131:281 — "the RPKM measure is inconsistent among samples"

Same data, two recipes: FPKM says $A > C$, TPM says $C > A$

axis, on the same board

- Your matrix has two axes. `.sum()` collapses one of them.
- `axis=0` collapses DOWN the rows → one number per COLUMN (per sample)
- `axis=1` collapses ACROSS the columns → one number per ROW (per gene)
- On our 3-gene × 2-sample board: `axis=0` prints 2 numbers, `axis=1` prints 3
- The TPM check sums down a column → `axis=0` → it must print one number per sample
- Mnemonic: the axis you name is the one that disappears

The same 3 × 2 board (rate = count ÷ kb). The axis you name is the one that disappears.

M.sum(axis=0)

	s1	s2
A	100	100
B	100	100
C	80	180
	280	380

down the rows → 2 numbers, one per SAMPLE
(axis 0, the gene axis, disappears)

M.sum(axis=1)

	s1	s2	
A	100	100	200
B	100	100	200
C	80	180	260

→ [200, 200, 260]

across the columns → 3 numbers, one per GENE
(axis 1, the sample axis, disappears)

M.sum(axis=0) → [280, 380]; M.sum(axis=1) → [200, 200, 260] — the axis you name disappears

Why differential expression is not run on TPM

- Within-sample question: which gene is more abundant in this tube? → TPM is fine
- Between-sample question: did this gene change? → TPM is not fine
- TPM's denominator is the sample's own pool — one change moves every other gene's TPM
- DE methods take raw counts and estimate their own size factors
- The rule, write it down: model raw counts; let the method do the normalisation
- Feeding TPM or FPKM to DESeq2 is the most common beginner error in RNA-seq

samples: want to see if differences across condition are significant (w.r.t. biological and technical variation)

features (e.g. genes)

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	679	448	873	408	1138
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	515	621	365	587
ENSG000000000457	260	211	263	164	245
ENSG000000000460	60	55	40	35	78

The input to DESeq2/edgeR is this: raw integer counts, genes × samples — RAW COUNTS ONLY

Median-of-ratios, in words

- For each gene: compute the geometric mean of its counts across all samples
- For each sample: take the ratio of each gene's count to that gene's geometric mean
- The sample's size factor is the median of those ratios
- The median is why a few enormous genes cannot move it — the composition correction
- Assumption: most genes are not changing. State it, because it is falsifiable
- Anders & Huber, Genome Biol 2010;11:R106 · TMM: Robinson & Oshlack, Genome Biol 2010;11:R25

Median-of-ratios (DESeq2): per gene a geometric mean, per sample a median

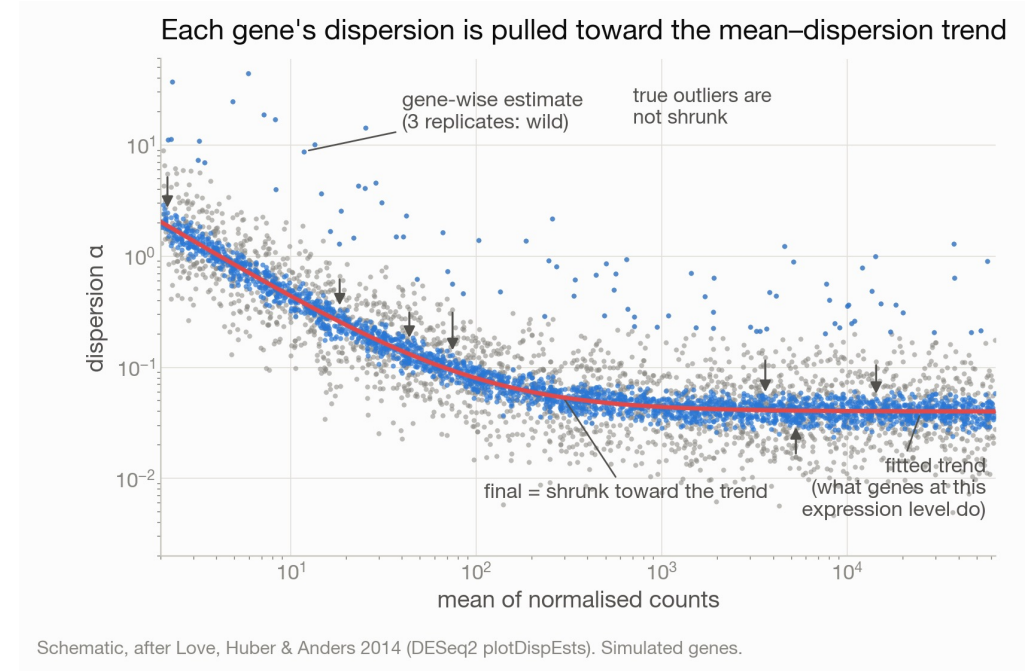
gene	count s1	count s2	count s3	geometric mean	s1 ÷ gm	s2 ÷ gm	s3 ÷ gm
ACTB	1,000	2,000	1,400	1,409	0.71	1.42	0.99
GAPDH	500	1,000	750	721	0.69	1.39	1.04
TP53	100	200	130	138	0.73	1.45	0.95
ALB (huge)	20,000	80,000	30,000	36,342	0.55	2.20	0.83
MYC	10	20	14	14	0.71	1.42	0.99
median of ratios = size factor					0.71	1.42	0.99
(mean of ratios, for contrast)					0.68	1.58	0.96

The single enormous gene (ALB, orange) drags the mean of the ratios; the median does not notice it. Assumption: most genes are not changing.
Anders & Huber, Genome Biol 2010;11:R106. Example numbers; same shape as the hbc training worked example.

Median-of-ratios worked: geometric mean per gene, ratios per sample, median = size factor (0.71, 1.42, 0.99)

The count model, in two sentences

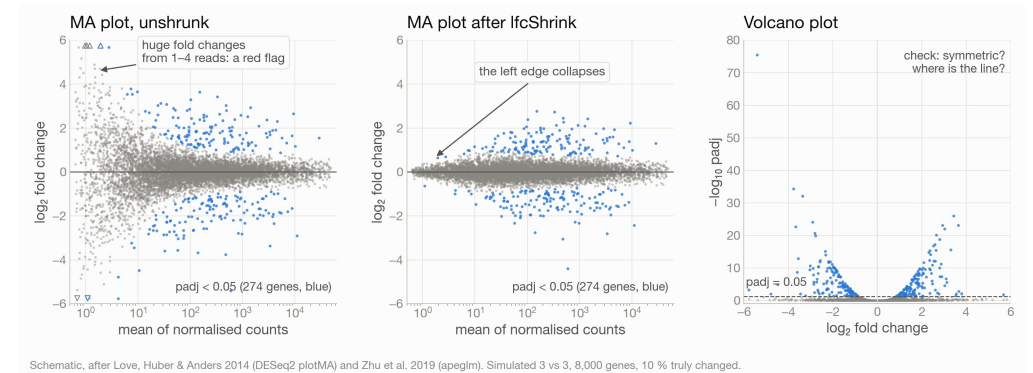
- Counts are not Poisson: replicates vary more than counting noise allows
- That extra variance is overdispersion
- DE methods use the negative binomial (negatif binom): a mean plus a per-gene dispersion (saçılım)
- With 3 replicates you cannot estimate 20,000 dispersions honestly
- So the method shrinks each gene's estimate toward a fitted mean–dispersion trend
- That is the whole idea. No derivation is examinable.
- Love, Huber & Anders, Genome Biol 2014;15:550 (DESeq2)



Schematic, after Love, Huber & Anders 2014: gene-wise dispersions (grey) shrunk toward the fitted trend (red) to final values (blue)

Reading a result: MA and volcano

- MA plot: $\log_2$ fold change (kat deęişim) on y, mean normalised count on x
 - the fan shape at low counts is real
- Volcano plot (volkan grafięi): $\log_2$ FC on x, $-\log_{10} p$ on y — check symmetry and the significance line
- lfcShrink pulls in the noisy fold changes of low-count genes
 - apegglm; Zhu et al., Bioinformatics 2019;35:2084
- Huge fold changes at the left edge of an unshrunk MA plot: a red flag, not a discovery

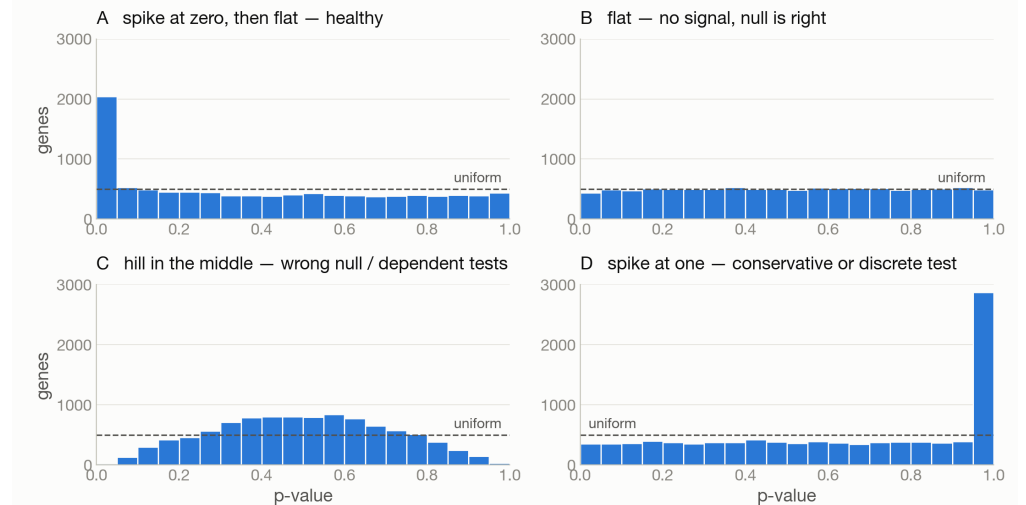


Schematic, after DESeq2 plotMA: unshrunk vs lfcShrink, and the matching volcano (simulated 3 vs 3)

pvalue is not padj, and only one of them may be filtered

- DESeq2 gives you both columns. They are not interchangeable.
- padj is already BH-corrected across all tested genes — filter on it freely
- Never filter on pvalue and then run BH yourself on the survivors — the Week 6 defect
- padj = NA is not missing data — the gene was filtered out or flagged as a count outlier
- Independent filtering: Bourgon, Gentleman & Huber, PNAS 2010;107:9546
- The p-value histogram from Week 6 is your five-second audit of anybody's DE output

Four p-value histograms — which one is a healthy differential-expression run?

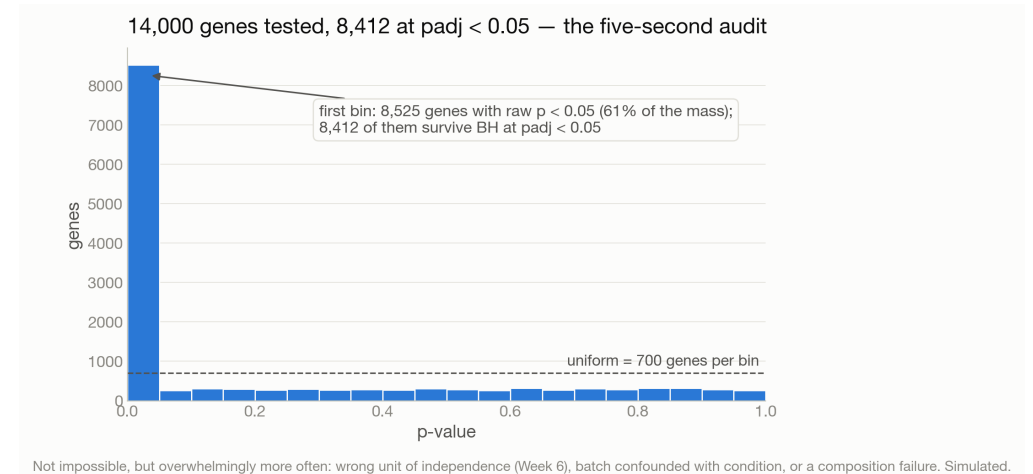


Schematic, after D. Robinson 2014, "How to interpret a p-value histogram". 10,000 simulated tests each.

Four p-value histograms, schematic after D. Robinson 2014: spike-at-zero (healthy), flat, hill, spike-at-one

ConcepTest: 8,412 of 14,000

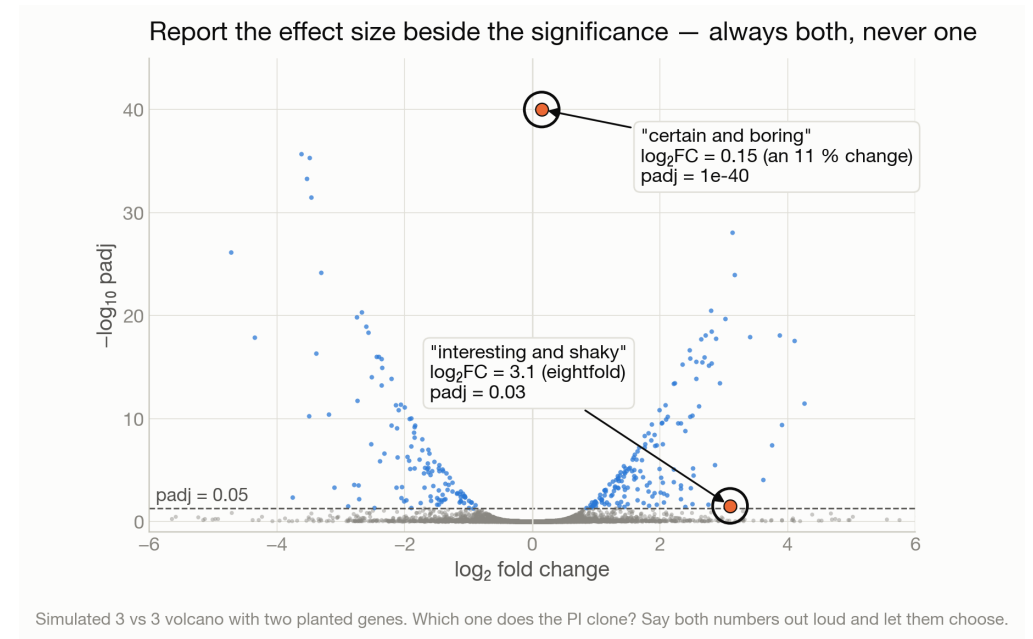
- A colleague runs DESeq2 on 6 vs 6 samples. 8,412 of 14,000 genes come back at $\text{padj} < 0.05$.
- Round 1 — vote alone: is this (a) a strong biological effect, (b) a pipeline problem, (c) impossible to say?
- Round 2 — turn to your neighbour, argue for 60 seconds, vote again
- Then: what is the ONE plot you ask them for before saying anything else?



The reveal: 14,000 genes, 8,525 with raw $p < 0.05$ (61% of the mass), 8,412 surviving BH at $\text{padj} < 0.05$

What a DE list licenses you to claim

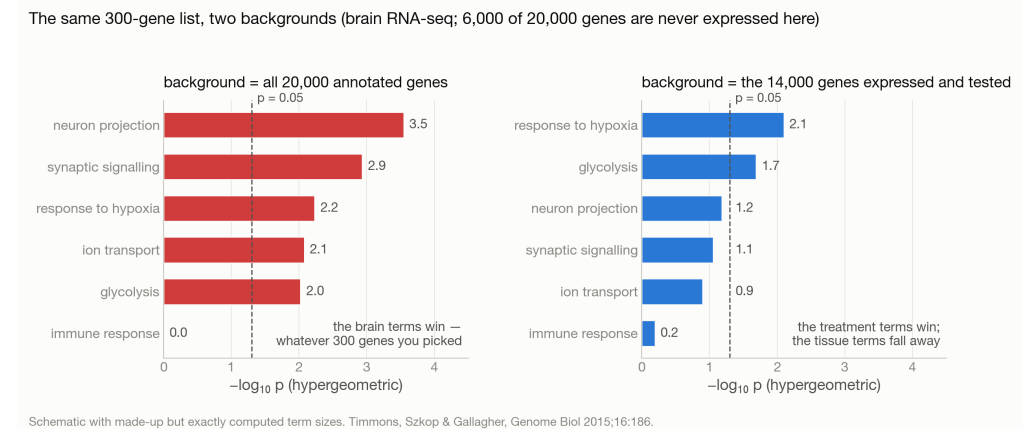
- A gene list is a hypothesis generator (hipotez üretici), not a mechanism
- Report the effect size (etki büyüklüğü) beside the significance — always both, never one
- $\text{padj} = 1\text{e-}40$ with a $\log_2$ fold change of 0.15 is a real, statistically certain, biologically uninteresting result
- "Significant" is a property of your sample size at least as much as of the biology
- The sentence you are allowed to write: "Gene X was higher in treated samples ($\log_2\text{FC} = 1.8$, $\text{padj} = 3\text{e-}6$, $n = 6$ donors)."
- The sentence you are not: "Gene X drives the treatment response."



'Certain and boring' ($\log_2\text{FC} 0.15$, $\text{padj} 1\text{e-}40$) vs 'interesting and shaky' ($\log_2\text{FC} 3.1$, $\text{padj} 0.03$)

Enrichment analysis, and its own trap

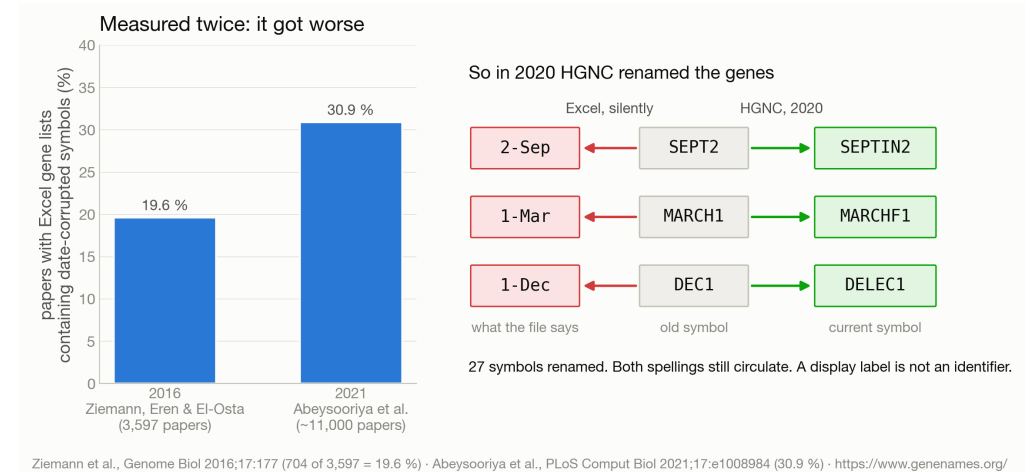
- You have 300 significant genes. Are they "about" anything? → GO / KEGG / Reactome
- The test asks: enriched for a pathway relative to a background (arka plan gen kümesi)
- The background is a choice, and it is usually made wrongly
- Wrong: all ~20,000 annotated genes. Right: the ~14,000 actually expressed and tested
- The wrong background manufactures enrichment for whatever your tissue expresses
- Timmons, Szkop & Gallagher, Genome Biol 2015;16:186 — "multiple sources of bias confound functional enrichment analysis"



Same 300 genes, two backgrounds: tissue terms top the 20,000-gene universe and fall below $p = 0.05$ with the 14,000 expressed genes

Entities: gene symbols are not identifiers — GENOMES E

- Excel silently converts SEPT2 → 2-Sep, MARCH1 → 1-Mar, DEC1 → 1-Dec
- Ziemann, Eren & El-Osta, Genome Biol 2016;17:177 — ~20% of papers with Excel supplements affected
- Abeysooriya et al., PLoS Comput Biol 2021;17:e1008984 — five years later, 30.9% of ~11,000 papers. It got worse.
- In 2020 HGNC renamed the genes: SEPT2→SEPTIN2, MARCH1→MARCF1, DEC1→DELEC1
- The community renamed its genes because a spreadsheet could not be trusted
- Both spellings still circulate in files people will hand you



Measured twice — 19.6% (2016) to 30.9% (2021) — and the 2020 HGNC renames

The identifiers you actually keep

- ENSG00000141510 — Ensembl gene ID: stable, versioned, never reused
- MANE Select — one Ensembl/RefSeq-agreed transcript per protein-coding gene
 - Morales et al., Nature 2022;604:310
- Symbol for humans; ENSG for machines to join on
- Never join two tables on gene symbol. Ever.
- GENOMES E row: three resolved identifiers

The table the lab makes you build — keep the symbol for humans, the ENSG for machines

what the collaborator's file says	current HGNC symbol	Ensembl gene ID (ENSG...)
2-Sep	SEPTIN2	
1-Mar	MARCHF1	
1-Dec	DELEC1	

The ENSG column is yours to fill from <https://www.genenames.org/> → Ensembl. Never join two tables on gene symbol.

File says → current HGNC symbol → Ensembl gene ID (yours to fill in the lab)